Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	XXXXXX
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability for the **Streamlining IT Procurement: Automating Standard Laptop Orders** project. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	**Yes**	Standard 4-space indentation throughout Python scripts.
2	Proper File Structure	Files and folders are logically organized	**Yes**	Modules structured for UI, API, database, and logic.
3	Meaningful Variable Names	Variables reflect their purpose clearly	**Yes**	Descriptive names like `laptop_model` & `employee_id`.
4	Function / Method Names	Functions are descriptively named	**Yes**	e.g., `submit_procurement_request()`.
5	Code Comments	Inline and block comments explain logic	**Yes**	Core workflow & error recovery steps documented.
6	Modular Design	Code is split into reusable functions/modules	**Yes**	Separated auth, catalog, order processing, & alerts.
7	No Redundant Code	Duplicate or unused code is removed	**Yes**	Shared HTTP client & database connection pool used.
8	Error Handling	Exceptions and errors are handled gracefully	**Yes**	Try-catch handlers for API calls & input validation.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	User Authentication Module	Python / OAuth 2.0	Employee Portal & Admin Dashboard	**High**
2	Notification Service	Python / SMTP API	Approval emails & IT dispatch alerts	**High**
3	Inventory Data Fetcher	SQL / REST API	Laptop stock management system	**Medium**
4	Procurement Form Validator	JavaScript / Python	Order creation & edit forms	**High**

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	**5**	Logical directory layout separating UI, logic, and external integrations.
Readability	**5**	Follows PEP 8 guidelines; consistent naming and formatting across files.
Reusability	**4**	Core notification, auth, and data layer components are modular and decoupled.
Documentation / Comments	**4**	Comprehensive block comments for workflow routines and function signatures.
Overall Score	**4.5 / 5**	**High-quality codebase designed for easy maintainability and future enhancement.**